

Jinning LIANG

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RESEARCH INTERESTS

Keywords: galaxy formation theory, galaxy-halo connection, AGN feedback, galactic dynamics, galactic chemical evolution, cosmology, numerical simulation, semi-analytical model

EDUCATION

The Kavli Institute for Astronomy and Astrophysics, Peking University <i>Ph.D. student in Astrophysics, Advisor: Prof. Fangzhou Jiang</i>	Sep 2025 - Sep 2029
The Institute for Computational Cosmology, Durham University <i>STFC-funded Ph.D. student in Physics, Advisor: Prof. Cedric Lacey</i>	Oct 2023 - Mar 2025
<i>M.Sc. in Physics received</i>	Oct 2023 - Sep 2024 Mar 2025
School of Physics and Technology, Wuhan University <i>Bachelor of Science in Physics</i>	Sep 2019 - Jun 2023
➤ Astronomy Class (Selected from pool of 280 students due to outstanding performance and enthusiasm for astronomy)	Sep 2020 - Jun 2023
➤ GPA: 3.80/4.00; 90.15/100; Ranking: 1/196	
The Kavli Institute for Astronomy and Astrophysics, Peking University <i>Visiting Student</i>	Mar 2023 - Oct 2024
Department of Astronomy, Peking University <i>CSST-Galaxies Observation Summer School Student</i>	Jul 2022
Shanghai Astronomical Observatory <i>Visiting Student</i>	Jan 2022 - Feb 2022

PUBLICATIONS (* Denotes co-first author, † Denotes corresponding author)

- [1] **Jinning Liang**, Lacey. C et al., *Implementation of Variable Coupling Efficiency Model in AGN feedback*, in preparation
- [2] **Jinning Liang**, F. Jiang[†], H.J. Mo, et al., *Connection between galaxy morphology and dark-matter halo structure II: predicting disk structure from dark-matter halo properties*, about to submit
- [3] F. Jiang^{†,*}, **Jinning Liang**^{*} et al., *Formation and Environmental Context of Giant Bulgeless Disk Galaxies in the Early Universe: Insights from Cosmological Simulations*, Submitted, 2025 [[2504.01070](#)]
- [4] Zou. S[†], Simcoe. R.A., Petitjean. P, Péroux. C, Champagne. J, Wang. F, **Jinning Liang** et al., *Disturbed cold gas in galaxy and structure formation*, Submitted, 2025 [[2502.14705](#)]
- [5] **Jinning Liang**, F. Jiang[†], H.J. Mo et al., *Connection between galaxy morphology and dark-matter halo structure I: a running threshold for thin discs and size predictors from the dark sector*, MNRAS 541 (2025) 2304 [[2403.14749](#)]
- [6] **Jinning Liang**, F. Jiang[†] et al., *Constrain the Dark Matter Distribution of Ultra-diffuse Galaxies with Globular-Cluster Mass Segregation: A Case Study with NGC5846-UDG1*, ApJ 964 (2024) 53 [[2304.14431](#)]
- [7] **Jinning Liang**, E. Gjergo[†] & X. Fan, *Assessing stellar yields in Galaxy chemical evolution: benchmark on observational stellar abundance patterns*, MNRAS 522 (2023) 863 [[2304.00208](#)]
- [8] E. Gjergo[†], A.G. Sorokin, A. Ruth, E. Spitoni, F. Matteucci, X. Fan, **Jinning Liang** et al., *GalCEM I - A Publicly-Available Detailed Isotopic Chemical Evolution Code*, ApJS 264 (2023) 44 [[2301.02257](#)]
- [9] H. Liu^{*}, **Jinning Liang**^{*} & J. Jia[†], *Deflection and Gravitational lensing of null and timelike signals in the Kiselev black hole spacetime in the weak field limit*, Class. Quantum. Grav 39 (2022) 195013 [[2204.04519](#)]

TEACHING

PHYS2631 Stars & Galaxies, Durham University <i>Workshop Demonstrator</i>	Oct 2023 - Apr 2024
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SCIENTIFIC OUTREACH

Celebrate Science, Durham University <i>Educated children with basic knowledge and illustration about gravitational lensing and galaxy formation</i>	2023
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TALKS (* Denotes invited talk)

[1] Conference Talk, The 3 rd Shanghai Assembly on Cosmology and Structure Formation, Shanghai, China <i>Connection between galaxy morphology and dark-matter-halo structure</i>	2025
[2] Conference Talk, CSST annual meeting, Kunming, China <i>Galaxy-halo coevolution: simulations and mock observations – Halo condition on disc formation</i>	2025
[3] Seminar Talk [*] , The Chinese University at Hongkong, Hongkong, China <i>Connection between galaxy morphology and dark matter halo structure – Halo condition on disc formation</i>	2025
[4] Seminar Talk, Flat Talk, Durham, UK <i>AGN feedback by line-driven winds</i>	2024

[5] Workshop Talk, 20 th Durham-Edinburgh eXtragalactic (DEX) Workshop, Durham, UK <i>New morphological decomposition method and varying circularity threshold</i>	2024
[6] Seminar Talk* , Shanghai Astronomical Observatory, Shanghai, China <i>Connection between galaxy morphology and dark-matter halo structure</i>	2023
[7] Seminar Talk* , Shanghai Jiaotong University, Shanghai, China <i>Dark matter properties and its connection with galaxy morphology</i>	2023
[8] Seminar Talk (online), University of Arizona, Arizona, AZ, USA <i>Globular Clusters in UDGs</i>	2022
[9] Conference Talk, ISM Physics and Chemistry Conference, Yichang, Hubei, China <i>Galactic stellar abundance scatter investigated through yield analysis in galaxy chemical evolution</i>	2022

SELECTED HONORS AND AWARDS

Science and Technology Facilities Council Scholarship		2023
Yu Gang - Song Xiao Scholarship of Wuhan University	45/30000	2022
First-class Scholarship of Wuhan University	Top 5%	2022
MCM&ICM Finalist Award	Top 2%	2022
National Astronomical Observatories Scholarship	3/600	2021

SKILLS

Programming Languages & Software: Python (Extensively), Mathematica (Extensively), Matlab and LaTeX
 Language: Mandarin (Native), English (Proficient)
 Simulations & Semi-analytical model: *IllustrisTNG*, *EAGLE*, *Swift-Colibre*, *NuPyCEE* and *SatGen*

MENTORSHIP

[1] Haiyang Xin , postgraduate at Peking University <i>Finding Milky Way catalog using galactic decomposition method (MorphDecom), in preparation</i>	since Jan 2024
[2] Peng Xu , undergraduate at Tsinghua University <i>Classifying and visualizing cosmic web with T-web method and DisPerSE, in preparation</i>	since Mar 2024
[3] Yalin Wu , undergraduate at Peking University <i>Simulating stellar streams and globular clusters in semi-analytical model, in preparation</i>	since Oct 2024
[4] Yiheng Tian , undergraduate at Peking University <i>Studying the evolution of bars quantities in IllustrisTNG, undergraduate thesis</i>	Dec 2024 - May 2025
[5] Jianyuan Luo , postgraduate at Peking University <i>Finding inner mini-disk of massive galaxies at high redshift using cosmological simulations, in preparation</i>	since Mar 2025

PROFESSIONAL REFERENCES

[1] Prof. Fangzhou Jiang <i>The Kavli Institute for Astronomy and Astrophysics, Peking University</i>	Email: fangzhou.jiang@pku.edu.cn
[2] Prof. Cedric Lacey <i>The Institute for Computational Cosmology, Durham University</i>	Email: cedric.lacey@durham.ac.uk
[3] Dr. Sownak Bose <i>The Institute for Computational Cosmology, Durham University</i>	Email: sownak.bose@durham.ac.uk
[4] Prof. Houjun Mo <i>Department of Astronomy, University of Massachusetts</i>	Email: hjmo@umass.edu
[5] Prof. Xilong Fan <i>School of Physics and Technology, Wuhan University</i>	Email: xilong.fan@whu.edu.cn